

Lesson 2 · Practice sheet

NCEA Level 2 · AS 91262 · Turning points and optimisation

Time: 30 minutes

Answer version · tutor's working, not an official mark scheme

Show all working. Every maximum or minimum needs a reason; every answer in context needs units. [A]
Achieved, [M] Merit, [E] Excellence.

1. Find the turning points of $y = x^3 - 27x + 10$ and justify their nature. [A]

$$\frac{dy}{dx} = 3x^2 - 27 = 0, \text{ so } x^2 = 9, x = 3 \text{ or } x = -3$$

$$x = 3: y = 27 - 81 + 10 = -44; \quad x = -3: y = -27 + 81 + 10 = 64$$

$$\frac{d^2y}{dx^2} = 6x: \text{ at } x = 3 \text{ it is } 18 > 0, \text{ so } (3, -44) \text{ is a minimum}$$

$$\text{at } x = -3 \text{ it is } -18 < 0, \text{ so } (-3, 64) \text{ is a maximum}$$

2. A stone is thrown upwards. Its height is $h = 1.5 + 14.7t - 4.9t^2$ metres after t seconds. Find its maximum height. [A]

$$\frac{dh}{dt} = 14.7 - 9.8t = 0, \text{ so } t = 1.5 \text{ s}$$

$$h = 1.5 + 14.7(1.5) - 4.9(1.5)^2 = 1.5 + 22.05 - 11.025 = 12.525$$

$$h'' = -9.8 < 0, \text{ so maximum. Maximum height } 12.5 \text{ m}$$

Question 3 [A]

- (c) Use calculus to find the values of x for which the graph of the function $f(x) = \frac{2x^3}{3} + \frac{3x^2}{2} - 20x - 3$ is decreasing.

$$f'(x) = 2x^2 + 3x - 20$$

$$\text{Decreasing where } f'(x) < 0: 2x^2 + 3x - 20 < 0$$

$$(2x - 5)(x + 4) < 0; f'(x) = 0 \text{ at } x = 2.5 \text{ and } x = -4$$

f' is an up-parabola, so it is negative between its roots:

$$f' \text{ is decreasing for } -4 < x < 2.5$$

Question 4 [A] · Merit with the justification

- (b) A cricket match is being played. A cricketer hits the ball, and its height, h , above the grass cricket pitch in metres, can be modelled by the function:

$$h = 22.5t - 4.9t^2 + 1$$

where t is the time in seconds since the ball was hit.

Using calculus methods, find the maximum height above the ground that the ball reaches.

$$h' = 22.5 - 9.8t = 0, \text{ so } t = 2.296 \text{ s}$$

$$h = 22.5(2.296) - 4.9(2.296)^2 + 1 = 26.83$$

$$h'' = -9.8 < 0, \text{ so this is a maximum}$$

Maximum height 26.8 m (above the grass)

Question 5 (i) [A] · (ii) [M]

- (c) The number of daily viewers for a new presenter on a video gaming streaming service can be modelled by the following equation:

$$V = -11t^2 + 528t \quad \{0 \leq t \leq 48\}$$

Where V represents the number of daily viewers and t represents time in months.

- (i) Find the rate(s) of change of daily viewers, with respect to t , when the number of daily viewers is 3520.

$$-11t^2 + 528t = 3520, \text{ so } t^2 - 48t + 320 = 0$$

$$(t - 8)(t - 40) = 0, \text{ so } t = 8 \text{ or } t = 40 \text{ months}$$

$$V' = -22t + 528$$

$$t = 8: V' = -176 + 528 = 352 \text{ viewers per month (increasing)}$$

$$t = 40: V' = -880 + 528 = -352 \text{ viewers per month (decreasing)}$$

- (ii) What is the maximum number of daily viewers that this new presenter gets during the first 48 months?

$$V' = -22t + 528 = 0, \text{ so } t = 24 \text{ months}$$

$$V = -11(24)^2 + 528(24) = 6336$$

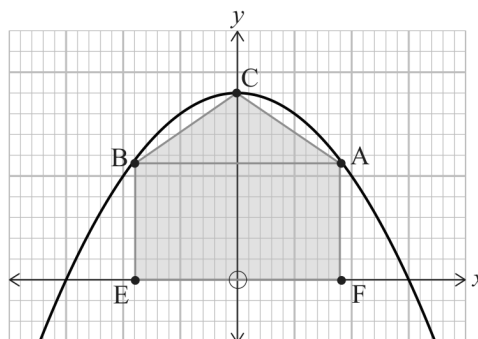
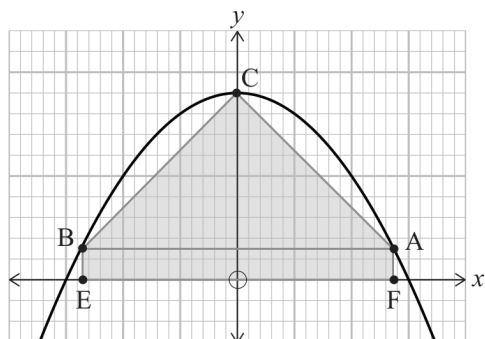
$$V'' = -22 < 0, \text{ so maximum: } 6336 \text{ daily viewers}$$

Question 6 [M] · Excellence with a full justification

(d) Two possible configurations of a house shape drawn below a parabola are shown below.

A house shape is drawn as shown where:

- C is at (0,9).
- A and B are points on the parabola $y = 9 - x^2$.
- E and F are on the x -axis.



(i) Find the height of the wall (AF or BE) when the area of the house shape is a maximum. Justify that this is the maximum area.

Let A be $(x, 9 - x^2)$, $x > 0$. Wall height $= 9 - x^2$

Rectangle ABEF: $2x(9 - x^2)$. Triangle ABC: base $2x$, height $9 - (9 - x^2) = x^2$

Area $= 2x(9 - x^2) + x \cdot x^2 = 18x - 2x^3 + x^3 = 18x - x^3$

$A' = 18 - 3x^2 = 0$, so $x^2 = 6$, $x = \sqrt{6} = 2.449$

Wall height $= 9 - 6 = 3$ units

$A'' = -6x = -14.7 < 0$ at $x = \sqrt{6}$, so the area is a maximum

(maximum area $= 12\sqrt{6} = 29.4$ square units)

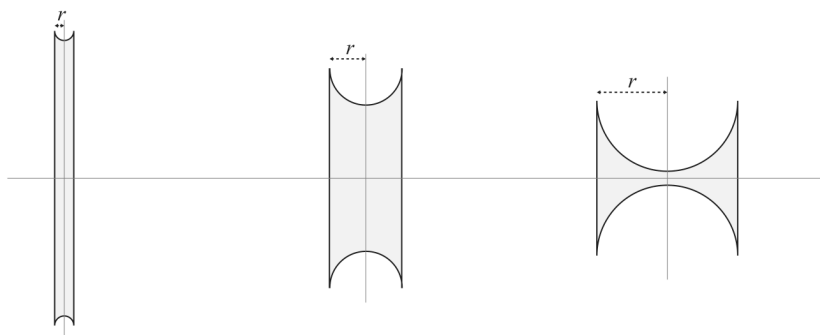
Question 7 Extension [E]

- (e) A new business is designing a logo based on a stylised 'H' shape, as seen in the diagrams.

The logo is to be designed with two semi-circles and two straight lines. The owner of the new business wants to include space for the company's name inside the shape, so he has asked the designer to maximise the area inside the shape.

The owner plans to build a replica of the logo, and due to material constraints, the total perimeter of the shape must not be greater than 80 cm.

Three examples are shown here:



- Find the maximum area.
- Use calculus methods to show that this is a maximum.

Circumference of a circle: $C = 2\pi r$

Area of a circle : $A = \pi r^2$

Straight lines length L ; two semicircles of radius r each have length πr

Perimeter: $2L + 2\pi r = 80$, so $L = 40 - \pi r$

Area = rectangle $2r \times L$ minus two semicircles = $2rL - \pi r^2$

$A = 2r(40 - \pi r) - \pi r^2 = 80r - 3\pi r^2$

(i) $A' = 80 - 6\pi r = 0$, so $r = \frac{40}{3\pi} = 4.244$ cm

$A = 80(4.244) - 3\pi(4.244)^2 = 169.8$ cm²

(ii) $A'' = -6\pi < 0$, so this is a maximum

Maximum area 169.8 cm² (when $r = 4.24$ cm, $L = 26.7$ cm)